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A Systematic Review of Intensive Empirically Supported Treatments for Posttraumatic Stress Disorder

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Various clinical practice guidelines for the treatment of posttraumatic stress disorder (PTSD) have consistently identified two frontline evidence-based psychotherapies (EBPs)—prolonged exposure (PE) and cognitive processing therapy (CPT)—as well as other empirically supported treatments (EST), such as eye movement desensitization and reprocessing (EMDR) and cognitive therapy for PTSD (CT for PTSD). However, researchers and clinicians continue to be concerned with rates of symptom improvement and patient dropout within these treatments. Recent attempts to address these issues have resulted in intensive, or "massed," treatments for PTSD. Due to variability among intensive treatments, including treatment delivery format, fidelity to the EST, and the population studied, we conducted a systematic review to summarize and integrate the literature on the impact of intensive treatments on PTSD symptoms. A review of four major databases, with no restrictions regarding publication date, yielded 11 studies that met all inclusion criteria. The individual study findings denoted a large impact of treatment on reduction of PTSD symptoms, *d*s = 1.15–2.93, and random-effects modeling revealed a large weighted mean effect of treatment, $d = 1.57$, 95% CI [1.24, 1.91]. Results from intensive treatments also noted high rates of treatment completion (i.e., 0%–13.6% dropout; 5.51% pooled dropout rate across studies). The findings suggest that intensive delivery of these treatments can be an effective alternative to standard delivery and contribute to improved treatment response and reduced treatment dropout.

Prevalence rates of posttraumatic stress disorder (PTSD) vary between civilian and veteran populations as well as across veteran service eras. Whereas the PTSD prevalence rate was found to range from 3.4% to 8.0% among a representative sample of civilians (Lehavot et al., 2018), higher rate estimates have been observed among veterans. For example, recent meta-analytic findings demonstrated a 23% rate of PTSD among a sample of recently returning veterans (Fulton et al., 2015).

Clinical practice guidelines for PTSD provide treatment recommendations for individuals diagnosed with PTSD and PTSD symptoms (e.g., Berliner et al., 2019; Courtois et al., 2017; Veterans' Affairs/Department of Defense [VA/DoD], 2017). In general, these guidelines recommend that individuals engage in individual, trauma-focused treatment in which the core of the treatment protocol involves exposure and/or cognitive restructuring (Berliner et al., 2019; Courtois et al., 2017; VA/DoD, 2017). Based on extensive research in both civilian and veteran samples, two evidence-based psychotherapies (EBPs) have been consistently recommended as frontline treatments: prolonged exposure therapy (PE) and cognitive processing therapy (CPT). Other therapies that have empirical support and have been recommended or suggested in these guidelines include eye movement desensitization and reprocessing (EMDR), cognitive therapy (CT) for PTSD, cognitive behavioral therapy (CBT), narrative exposure therapy (NET), brief eclectic therapy (BET), and written exposure therapy (WET; Berliner et al., 2019; Courtois et al., 2017; VA/DoD, 2017).

Each of these treatments—henceforth referred to as empirically supported treatments (ESTs)—have demonstrated effectiveness in the treatment of PTSD among civilians, military personnel, and/or veterans (e.g., Steenkamp et al., 2015; Watts et al., 2013). However, not all individuals achieve clinically significant reductions in PTSD symptoms through treatment (Bradley et al., 2005; Resick et al., 2002; Schottenbauer et al., 2008; Steenkamp et al., 2015). Efforts to address treatment nonresponse have examined session frequency as a potential mechanism to enhance EST delivery and effectiveness. For

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instance, more frequent sessions of PE and CPT, referred to as intensive or "massed" treatments (hereafter referred to as intensive treatments), have been found to result in improved PTSD treatment outcomes (Gutner et al., 2016; Wachen et al., 2019). Specifically, fewer days between sessions have been shown to lead to higher levels of symptom amelioration throughout the course of treatment; in contrast, the longer a delay between sessions following introduction to exposure assignments, the smaller the reduction in PTSD symptoms (Gutner et al., 2016). These results emphasize the importance of session frequency on treatment outcomes.

Session frequency may also influence treatment response through a reduction in patient dropout. Patient dropout is a major concern in both trauma-focused EST research and clinical practice (e.g., Garcia et al., 2011; Hembree et al., 2003; Imel et al., 2013; Kehle-Forbes et al., 2016; Najavits, 2015; van Minnen et al., 2002; Zayfert et al., 2005). One meta-analysis reported an average dropout rate of 18% across trauma-focused interventions for PTSD (Imel et al., 2013); however, more recent studies have identified dropout rates of between 30% and 39% (Berke et al., 2019; Eftekhari et al., 2019; Steenkamp et al., 2015). Higher rates of dropout from ESTs for PTSD, such as PE, have been observed in clinical settings (e.g., 28%–30%; Eftekhari et al., 2019; Zayfert et al., 2005) compared to controlled study designs (e.g., 20.5%; Hembree et al., 2003). Moreover, study results have shown significant variability in treatment dropout across trauma-focused interventions and population types (Schottenbauer et al., 2008).

Although treatment dropout is complex in both clinical and research settings, research has supported length of time between sessions as one, among many, relevant factors that influence dropout. Studies of intensive ESTs have utilized various treatment delivery schedules. For the present paper, intensive treatment was operationalized to refer to treatment delivered more than twice weekly. This definition was chosen for two primary reasons. First, past research on intensive treatment for anxiety disorders has examined treatment delivered twice weekly as a comparison condition to intensive intervention (e.g., Abramowitz et al., 2003). Second, many randomized control trials (RCTs) of standard delivery deliver ESTs twice weekly (e.g., Gutner et al., 2016).

The effectiveness of intensive ESTs for PTSD is an emerging area of research (see Wachen et al., 2019, for a narrative review); however, these treatments have already been examined across various populations, including veterans, adolescents, and civilians; trauma types, such as combat exposure, childhood sexual abuse, and interpersonal trauma (Ehlers et al., 2014; Foa et al., 2018; Hendriks et al., 2018; Jarero et al., 2015); and formats, including standalone delivery of a standalone EST, combinations of ESTs and adjunctive interventions (Foa et al., 2018; Zalta et al., 2018). Intensive ESTs for PTSD have also been implemented in several treatment settings and across levels of care (e.g., Norman et al., 2016; Wagenmans et al., 2018).

The results of studies that have examined intensive ESTs for PTSD are promising regarding treatment response (e.g., high rates of symptom amelioration; Bryan et al., 2018; Zoet et al., 2018) and reductions in treatment dropout (e.g., Bryan et al., 2018; Harvey et al., 2018; Wagenmans et al., 2018) when compared to research of standard delivery formats. Therefore, the aim of the present study was to conduct a systematic review of the literature and synthesize the existing data on this topic to explore the potential benefits of increasing session frequency when treating PTSD.

Method

Search Strategy

We conducted a systematic review of the literature across four major databases: PubMed, PsycINFO, PTSDpubs (formerly PILOTS), and GoogleScholar. The first two authors independently conducted searches of all articles published through October 2019. The search was not restricted based on publication year nor publication in a peer-reviewed journal, and GoogleScholar was utilized to capture unpublished works. Full-text searching was used when possible. Search terms encompassed three themes: treatment intensity, treatment type, and PTSD diagnosis. For the treatment intensity theme, search terms included "intensive" or "massed." For the treatment type theme, search terms included "prolonged exposure," "PE," "cognitive processing therapy," "CPT," "eye movement desensitization and reprocessing," "EMDR," "cognitive therapy," "CT," "cognitive behavioral therapy, "CBT," "narrative exposure" therapy," "NET," "brief eclectic therapy," "BET," "written exposure therapy," or "WET." For the PTSD diagnostic status theme, search terms included "posttraumatic stress disorder" or "PTSD." Due to the number of treatments included in the search, some of the databases we used could not accommodate the search in a single string of search terms. Therefore, searches for each treatment type were conducted independently in each database using a Boolean technique. That is, to capture articles that used intensive prolonged exposure for PTSD, the search string was: (intensive OR massed) AND ("posttraumatic stress disorder" OR PTSD) AND ("prolonged exposure" OR PE). Furthermore, as additional checkpoints, we reviewed the reference sections of empirical studies and narrative reviews that examined intensive treatments for PTSD (e.g., Wachen et al., 2019).

Study Selection

The criteria for inclusion were: (a) an empirical study (e.g., uncontrolled clinical trial, RCT) that examined an EST for PTSD (e.g., PE, CPT, EMDR) that was delivered in an intensive format; (b) study assessed treatment effectiveness in adults (18 years of age and older) who had a diagnosis of PTSD or probable PTSD, regardless of trauma type, via a clinician-administered interview; and (c) treatment delivery was provided in an outpatient setting. Finally, although we aimed to examine patients at the outpatient level of care, programs that offered lodging for participants' convenience or to those who

were traveling were included. These studies were permitted because the programs were intended to be outpatient programs.

Studies were excluded if they (a) included a primary intervention not identified by the clinical practice guidelines as an EST for PTSD (e.g., hypnosis, art therapy), (b) were a case report or case series, or (c) did not clearly delineate the nature or frequency of treatment. Studies were also excluded if (d) participants included children or adolescents, (e) participants were not formally assessed and diagnosed with PTSD via clinician or clinical interview (e.g., PTSD assessed via a self-report measure, only; Jarero et al., 2015), or (f) treatment was delivered in a residential or inpatient setting. These exclusion criteria were selected based on past research that has found differences between veterans receiving PTSD treatment in outpatient and residential settings with regard to demographic variables and pretreatment symptom severity (Haller et al., 2016; Walter et al., 2014), level of impairment (Haller et al., 2016), and the average number of problems to be addressed in treatment (Rosen et al., 2013). Moreover, residential and inpatient treatment settings often include milieu therapy, making it difficult to attribute symptom change or dropout rates to a specific EST offered within this type of setting.

Data Extraction and Quality Assessment of Studies

Two authors (NAS and AJW) extracted relevant data from the full text of all included studies. Analyses for the current study included calculating a weighted mean effect size estimate and pooled dropout rate across studies. Therefore, relevant data included pre- and post-PTSD symptom severity scores, sample size, and the number of treatment dropouts. Study quality for each of the articles was then assessed using an existing evidence appraisal checklist for single-study interventions with minor adaptations, which resulted in a 16-item checklist evaluating study validity (e.g., "Were the instruments used to measure the outcomes valid and reliable?"), reliability (e.g., "Did the study have a sufficiently large sample size?"), and applicability (e.g., "Can the results be applied to my population of interest?;" Cincinnati Children's Hospital Medical Center, 2012). Each item was scored on a scale that included options of 0 (*no*), 1 (*possibly, but not enough information is available*), and 2 (*yes*), resulting in a possible total score of 32 points.

Results

Search Results and Summary of Studies

The identification and inclusion of relevant studies are presented in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Moher et al., 2009) in Figure 1. To ensure accuracy and reduce coder bias, the first two authors independently searched, screened, and coded all articles. The initial review yielded a total of 408 articles. Duplicate articles were first eliminated, and the remaining articles were assessed based on title and abstract. Full-text publications were obtained for studies that met the inclusion criteria based on this information as well as studies for which a determination could not be made based on the abstract. This resulted in a total of 13 publications that met all inclusion criteria. There was a 100% independent-rater agreement before discussing search results.

Corresponding authors were contacted to determine whether publications utilized a unique sample population or a sample population that overlapped with another publication by the same group of authors. Based on correspondence with authors, two articles were removed due to confirmed or probable overlapping samples. In these instances, the article with the larger sample size was retained. As a result, we included 11 articles in the present review. Finally, corresponding authors were contacted if additional information was required for calculation of the weighted mean effect size estimate or dropout rate; this information was not obtained for one article (Foa et al., 2018).

The heterogeneity of study design and ESTs employed across the 11 studies that met inclusion criteria is noteworthy (see Tables 1 and 2 for individual study details). For instance, articles that met the inclusion criteria incorporated the use of a standalone EST, such as PE (Foa et al., 2018; Hurley, 2018); a combination of ESTs, such as PE and EMDR (van Woudenberg et al., 2018; Wagenmans et al., 2018; Zoet et al., 2018); or a combination of an EST and an adjunctive intervention, such as CPT and mindfulness skills (Zalta et al., 2018). Some studies also aimed to reduce barriers to care (e.g., distance from the facility) by offering lodging (e.g., Hurley, 2018; Wagenmans et al., 2018).

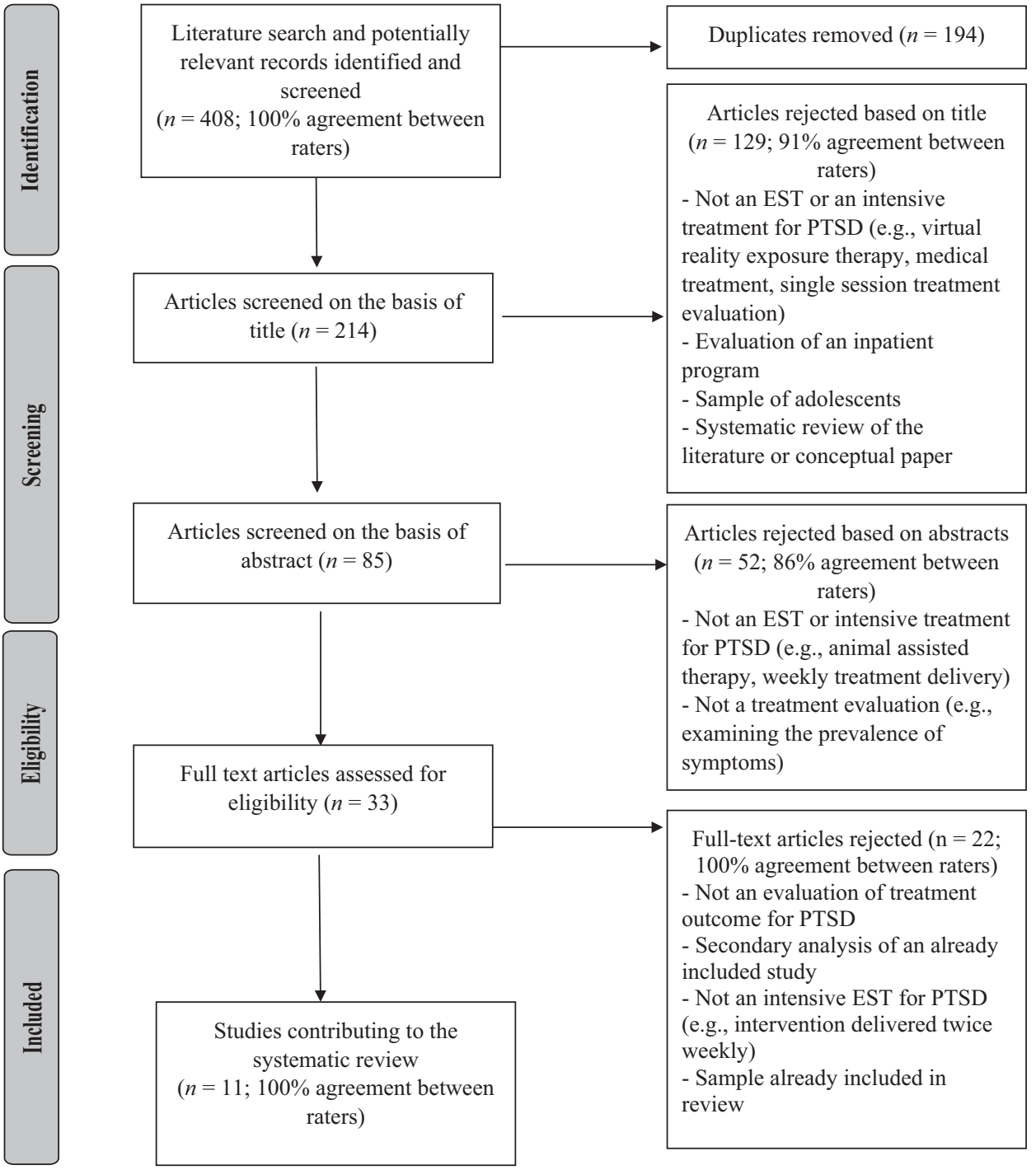
Study Quality

Quality ratings were conducted independently by the first two authors; there was a 97% rater agreement. Discrepancies among ratings were discussed until a consensus was reached. Based on our quality checklist, the average score for the articles included in this review was 26.4 out of 32 (range: 22–32). Of note, most studies addressed the impact of sample size on study results or conducted a power analysis to some degree (72.7%; $n = 8$), assessed or discussed adverse events that occurred during intensive treatment (54.5%; $n = 6$), and included follow-up assessments (54.5%; $n = 6$). Two studies were RCTs (18.2%), and all others were uncontrolled studies. A table with the quality assessment items and individual study scores can be found in the Supplemental Materials.

Symptom Amelioration

The findings reported in the included studies revealed a clinically significant reduction in PTSD symptoms, with a large magnitude of effect, Cohen's ds = 1.15–2.93 (Figure 2). To examine the overall impact of treatment on symptom reduction, a weighted mean effect size was calculated using the MetaEssentials program (Suurmond et al., 2017). Due to its ability to assign more balanced weights across studies rather than a fixed weight based on sample size alone, random-effects modeling

Figure 1
Flowchart of Relevant Study Identification



Note. EST = empirically supported treatment; PTSD = posttraumatic stress disorder.

was selected to calculate the mean estimate of treatment effect. The random-effects model was also selected because it is unlikely, due to the variability in study designs, that all the studies were functionally equivalent (Borenstein et al., 2007). One study was excluded from this calculation due to unavailable data (i.e., the standard deviation of mean PTSD symptom scores; Foa et al., 2018). The results of the random-effects modeling revealed a mean Cohen's *d* effect size 1.57, 95% CI [1.24, 1.91], which denotes a large impact of treatment on symptom reduction.

Table 1
Overview of the Included Studies

First author (year)	Study design	Review classification	Population and/or trauma type	*N*	Intervention(s)	Intervention frequency
Bryan (2018)	Uncontrolled trial	Combination treatment	Service members and veterans (e.g., combat trauma, sexual trauma)	20	CPT + adjunctive intervention (e.g., sports)	Daily individual CPT (without account)
Ehlers (2010)	Uncontrolled trial	Standalone EST	Accident, interpersonal violence	14	CT for PTSD	18 hr of therapy over 5–7 business days; 2 sessions daily for 90 min – 2 hr/session
Ehlers (2014)	RCT	Standalone EST	Interpersonal violence; accident/disaster; death/harm to others; other	30	CT for PTSD	18 hr of therapy delivered over 5–7 working days
Foa (2018)	RCT	Standalone EST	Service members and veterans (e.g., combat-related trauma)	110	Massed PE vs. spaced PE vs. PCT vs. MCC	Massed PE (daily sessions over 10 consecutive weekdays); spaced PE (10 sessions over 8 weeks); PCT was delivered the same as spaced PE; MCC received 10–15-min phone calls 1 time per week for 4 weeks
Harvey (2018)	Uncontrolled trial	Combination treatment	Service members and veterans (e.g., deployment-related trauma)	111	PE or CPT + adjunctive interventions (e.g., DBT)	Daily individual PE or CPT (60 min) over the course of a 2-week day-treatment program
Hendriks (2018)	Uncontrolled trial	Standalone EST	Adults with interpersonal trauma (e.g., sexual abuse, physical abuse)	73	PE	Intensive phase involved 3 90–min sessions daily for 4 treatment days within 1 week; after the intensive phase, there were 4 weekly 90-min PE booster sessions with HW

(*Continued*)

Table 1
Continued

First author (year)	Study design	Review classification	Population and/or trauma type	*N*	Intervention(s)	Intervention frequency
Hurley (2018)	Uncontrolled trial; secondary analysis	Standalone EST	Service members and veterans (e.g., combat-related trauma, medical trauma)	15	EMDR vs. weekly EMDR	Daily EMDR 2 times/day (75 min) during a 10-day period vs. weekly EMDR (50–60 min); both formats provided 18–20 total sessions
Wagenmans (2018)	Uncontrolled trial	Combination treatment	Adults with multiple trauma types (e.g., sexual abuse, physical abuse, war)	165	PE + EMDR (both with slight modifications) + adjunctive intervention (e.g., physical activity)	Two sets of 4 consecutive days of treatment with 3 days off between; 1 PE session and 1 EMDR session daily (3 hr of individual therapy daily)
Yasinski (2018)	Uncontrolled trial; Combination treatment		Military combat, MST	49	PE + adjunctive interventions (e.g., yoga)	Daily 90 min imaginal session and 120 min group in vivo exposure session
Zalta (2018)	Uncontrolled trial	Combination treatment	Service members and veterans (e.g., combat-related trauma, MST)	191	CPT + adjunctive interventions (e.g., yoga)	Daily individual + group CPT for 15 days; treatment included 15 individual and 13 group sessions
Zoet (2018)	Uncontrolled trial	Combination treatment	Adults with multiple trauma types (e.g., sexual abuse, physical abuse)	168	PE + EMDR (both with slight modifications) + adjunctive intervention (e.g., physical activity)	Two sets of 4 consecutive days of treatment; 1 PE session (90 min) and 1 EMDR session (90 min) daily (3 hr of individual therapy daily)

Note. ESTs = Empirically supported treatments; CPT = cognitive processing therapy; PE = prolonged exposure; DBT = dialectical behavioral therapy; EMDR = eye movement desensitization and reprocessing; MST = military sexual trauma; CT = cognitive therapy; RCT = randomized control trial; PCT = present-centered therapy; MCC = minimal contact control.

Table 2
Overview of the Variables of Interest, by Study

First author (year)	PTSD outcome measure(s)	PTSD diagnosis posttreatment	Follow-up	Results at follow-up	Boarding available
Bryan (2018)	CAPS-5, PCL-5	25% met criteria for PTSD at posttreatment, 43.4% met criteria at follow-up	6 months	Statistically significant increase in CAPS-5 scores; no significant change in PCL scores	Yes
Ehlers (2010)	CAPS, PDS, Sheehan Disability Scale, BAI, BDI	71.4% no longer met criteria for PTSD at 3 weeks posttreatment, 85.7% at 3 months posttreatment, and 92.9% at 9 months posttreatment	3 and 9 months	Participants in intensive condition had comparable improvement to those in a weekly comparison group	No
Ehlers (2014)	CAPS, PDS, Sheehan Disability Scale, BAI, BDI, QLES	73.3% and 66.7% lost PTSD diagnosis at posttreatment and 40-week follow-up compared to 77.4% and 74.2% in standard condition	27 and 40 weeks	Higher rates of assessor-rated remission in standard condition at 27 weeks (67.6%) and 40 (74.2%) weeks vs. intensive condition (40% and 53.3%, respectively)	No
Foa (2018)	PSS-I, PCL-S	Decreased rate of PTSD diagnosis at posttreatment (54.5%); this increased at 6-month follow-up (60.3%)	2 weeks, 12 weeks, 6 months	Massed PE resulted in lower symptom severity than MCC at 2-week follow-up; massed PE effects maintained at 12-week follow up, comparable to spaced PE	No

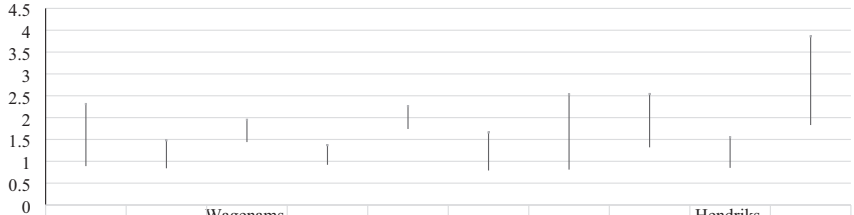
(*Continued*)

Table 2
Continued

First author (year)	PTSD outcome measure(s)	PTSD diagnosis posttreatment	Follow-up	Results at follow-up	Boarding available
Harvey (2018)	PCL-5		None	N/A	
Hendriks (2018)	CAPS		6 months	Participants maintained treatment gains over time; authors report this is noteworthy as the sample had previously failed outpatient treatment and presented with multiple childhood trauma exposure	No
Hurley (2018)	PCL-5, PCL-M, IES		1 year	Symptom improvement was maintained at 1-year follow-up, with a trend toward further PTSD symptom reduction	Yes
Wagenmans (2018)	CAPS, PSS-SR, ITEC, IES		None	N/A	No
Yaskinki (2018)	CAPS, MINI	47% positive for PTSD at completion, down from 90% at baseline	None	N/A	Yes
Zalta (2018)	CAPS-5, PCL-5	53.4% no longer met criteria for probable PTSD based on past-week PCL-5 scores	None	N/A	Yes
Zoet (2018)	CAPS, PSS-SR, ITEC, IES	84.1% showed treatment response, 55.3% lost PTSD diagnosis, 27.1% were in remission	None	N/A	No

Note. ESTs = empirically supported treatments; CAPS-5 = Clinician-Administered PTSD Scale for *DSM-5*; PCL-5 = PTSD Checklist for *DSM-5*; CAPS-IV = Clinician-Administered PTSD Scale for *DSM-IV*; ITEC = Interview for Traumatic Events in Childhood; IES = Impact of Events Scale; IES-R = Impact of Events Scale–Revised; PCL-M = PTSD Checklist–Military Version; PCL-S = PTSD Checklist–Stressor-Specific; PSS-I = PTSD Symptom Scale–Interview; PSS-SR = PTSD Symptom Scale–Self-Report.

Figure 2
Cohen's d Effect Size Estimates, With 95% Confidence Intervals, of Intensive Empirically Supported Treatments for Posttraumatic Stress Disorder (PTSD).



	Bryan et al. (2018)	Harvey et al. (2018)	Wagenams et al. (2018)	Zalta et al. (2018)	Zoet et al. (2018)	Yasinski et al. (2018)	Ehlers et al. (2010)	Ehlers et al. (2014)	Hendriks et al. (2018)	Hurley (2018)
Cohen's d	1.63	1.16	1.7	1.15	2.01	1.24	1.72	1.95	1.21	2.93
Lower CI	0.89	0.84	1.44	0.92	1.74	0.79	0.81	1.32	0.85	1.83
Upper CI	2.31	1.48	1.95	1.37	2.26	1.66	2.53	2.54	1.55	3.86

Note. Effect sizes were calculated using pre–posttreatment data, when available; otherwise, follow-up data were supplemented. Confidence intervals were not calculated if pre–posttreatment means or standard deviations were not included in the original study and corresponding authors could not be reached.

Moreover, the findings from studies in which long-term follow-up assessments were conducted found that intensive treatments were associated with large improvements in PTSD symptoms at varying posttreatment intervals (Bryan et al., 2018; Ehlers et al., 2010, 2014; Foa et al., 2018; Hendriks et al., 2018; Hurley, 2018; Van Woudenberg et al., 2018), Cohen's *d*s = 0.74–2.59.

Treatment Dropout

Five of the 11 identified studies examined a standalone EST intervention delivered in an intensive format: two studies utilized PE, one study utilized EMDR, and two studies utilized CT for PTSD. These five studies varied in study design, population, location of services, and EST. Despite this variability, all studies with an available comparison group reported a lower or similar dropout rate from the intensive condition compared with the standard treatment condition (Ehlers et al., 2014; Foa et al., 2018; Hurley, 2018). Of note, the study that reported the highest dropout rate (n = 15, 13.6%) from an intensive treatment condition (i.e., daily PE; Foa et al., 2018) included participants who never initiated treatment; of the participants who initiated therapy, 3.6% ($n = 4$) did not complete treatment. The reported dropout rate of 13.6% was used when calculating the pooled dropout rate across studies because these participants were randomized to the intensive condition. Additionally, each of the included studies reported a lower dropout rate than what has been observed in meta-analytic studies examining treatment dropout in ESTs for PTSD (Imel et al., 2013). Across these five studies, the dropout rate ranged from 0% to 13.6% (i.e., Ehlers et al., 2010, 2014; Foa et al., 2018; Hendriks et al., 2018; Hurley, 2018). Low dropout rates were also reported among studies that examined combinations of ESTs and adjunctive interventions (range: 0%-7.9%; e.g., Bryan et al., 2018; Zalta et al., 2018).

A pooled dropout rate was calculated by summing both the total number of participants who dropped out of treatment and the total number of participants enrolled in each of the included studies and determining the proportion of participants who did not complete treatment (see Table 3). The pooled dropout rate indicated that 5.51% of participants dropped out of the intensive intervention across studies, a rate that is substantially lower than what has been reported in research on standard delivery of trauma-focused treatment (e.g., range: 18%–39%; Berke et al., 2019; Eftekhari et al., 2019; Hembree et al., 2003; Imel et al., 2013; Steenkamp et al., 2015; Zayfert et al., 2005).

Discussion

The purpose of the present review was to assess the association between intensive ESTs for PTSD in an outpatient setting on symptom reduction and treatment dropout. We specifically focused on ESTs consistent with the clinical practice guidelines for PTSD (Berliner et al., 2019; Courtois et al., 2017; VA/DoD, 2017). Individual study quality assessments highlighted the range in study quality, suggesting that findings should be interpreted with caution, particularly regarding symptom amelioration, as 81.2% (n = 9) of the studies were uncontrolled. As a result, it is difficult to determine whether the symptom improvement demonstrated in most of the included studies was related to the intervention delivered or other factors, such as time, natural recovery, or regression to the mean.

Across the included studies, participants experienced a significant reduction in PTSD symptoms, and many participants no longer met the full criteria for PTSD posttreatment. Although the assessment of study quality revealed that follow-up assessments were not consistently employed (i.e., they were present in 54.5% of included studies; n = 6) and were administered at varying timepoints posttreatment, maintenance of treatment gains was also supported such that participants' posttreatment symptom levels remained lower than pretreatment levels in the studies that utilized follow-up data (e.g., Bryan et al., 2018; Foa et al., 2018; Hendriks et al., 2018). Future research should regularly implement follow-up assessments to further examine the maintenance of symptom change. These findings suggest that intensive treatment delivery may contribute to both

Table 3
Number of Dropouts in Each Study and Pooled Dropout Percentage

First author (Year)	Type of EST	Intensive condition (*n*)	Dropout (*n*)	Dropout (%)
Bryan (2018)	CPT	20	0	0
Ehlers (2010)	CT for PTSD	14	0	0
Ehlers (2014)	CT for PTSD	30	1	3.3
Foa (2018)	PE	110	15	13.6
Harvey (2018)	PE or CPT	88	4	4.5
Hendriks (2018)	PE	73	4	5.5
Hurley (2018)	EMDR	15	0	0
Wagenmans (2018)	PE + EMDR	165	7	4.2
Yasinski (2018)	PE	49	1	2
Zalta (2018)	CPT	176	15	8.5
Zoet (2018)	PE + EMDR	168	3	1.8
Pooled dropout[a]				5.51

Note. CPT = cognitive processing therapy; PTSD = posttraumatic stress disorder; EMDR = eye movement desensitization and reprocessing; PE = prolonged exposure.
[a]The number of participants included in the pooled dropout rate calculation for randomized controlled trials or quasi-experimental studies included participants in the intensive treatment condition. For studies that had multiple tracks (i.e., Harvey et al., 2018), only the PTSD track was included in this calculation.

short- and long-term symptom change and reduced treatment dropout among individuals diagnosed with PTSD.

The results of the included RCTs (Ehlers et al., 2014; Foa et al., 2018) and the nature of the compressed treatment delivery of intensive interventions suggest that this delivery format may provide more rapid symptom relief compared to standard treatment. This is due in part to the full protocol being delivered within a shorter timeframe than the standard delivery. For example, posttreatment assessment for an intensive EST may occur after a few weeks versus a few months with standard delivery. One primary advantage of more rapid symptom relief is the ability to improve overall functioning more rapidly, which facilitates engagement in or return to previous life activities sooner than with traditional approaches. In addition to more rapid symptom reduction, intensive treatment was associated with symptom improvement in a variety of populations and trauma types across the included studies. For instance, a combination of PE and EMDR was associated with symptom improvement among childhood abuse survivors (e.g., Wagenmans et al., 2018), a population in which trauma-focused treatment is typically addressed using a phase-oriented approach (e.g., Herman, 1997). Additional RCTs are needed to determine the effect of intensive treatment on symptom amelioration.

Finally, the present findings revealed a low pooled dropout rate across studies, which was less than the average dropout rate reported for standard ESTs for PTSD (e.g., Imel et al., 2013). The differences in treatment dropout between intensive and standard EST delivery are further supported by the results from the RCTs and quasi-experimental studies included in this review (e.g., Ehlers et al., 2014; Foa et al., 2018; Hurley, 2018). This low dropout rate may improve the extent to which individuals with PTSD receive the minimum therapeutic dose of an intervention and, thus, further facilitate symptom reduction. It is important to note that other factors may be associated with treatment dropout, such as motivation for treatment and therapeutic alliance. As a result, future research should assess the influence of these factors among individuals who engage in traditional versus intensive treatments for PTSD, particularly in uncontrolled studies.

Although research in this area is preliminary and ongoing, taken together, the results of this review suggest that the use of intensive interventions may be a viable alternative to standard delivery of ESTs for PTSD. Intensive ESTs may facilitate individuals in successfully achieving symptom reduction and completing a trauma-focused intervention. This symptom improvement could also impact buy-in for treatment, perhaps increase the likelihood an individual will seek services in the future if needed. Of particular note is the range and frequency of comorbidities and trauma types represented in several of the included studies. By including individuals with co-occurring disorders and a range of trauma exposures, these findings provide preliminary evidence to support the generalizability of the results.

Due to the nascent state of the literature on intensive ESTs for PTSD, few RCTs have been conducted to date. As noted previously, the limited number of RCTs renders drawing conclusions about the effect of intensive treatments on symptom change over time tenuous. Therefore, ascribing changes to the frequency of the intervention cannot be definitively stated at this time because other, uncontrolled factors may be responsible. This is a primary limitation of the current study, as other factors may better account for the effect of treatment.

Similarly, heterogeneity across studies is important when drawing conclusions about the association between intensive treatments, PTSD symptom change, and treatment dropout. Future research could benefit from a more thorough examination of standalone ESTs, delivered without the incorporation of additional services, as including adjunctive interventions may require procuring additional resources, such as providers, and

may augment symptom change. Moreover, dismantling existing combination-treatment protocols, such as an EST delivered with yoga or mindfulness sessions, could determine specific factors that drive symptom change within these programs. For instance, whereas the intensive EST could reduce trauma-related symptoms, the combination of yoga and mindfulness may serve as a form of behavioral activation, which may augment the reduction of depressive symptoms and improve distress tolerance, thereby further impacting trauma-related symptom change. Such future research may also identify moderator variables and illuminate populations or trauma types that may be better suited for a combination of therapies rather than a standalone EST. Additional study variability included whether boarding or lodging was offered to study participants (e.g., Bryan et al., 2018; van Woudenberg et al., 2018). Programs that offer lodging or boarding could contribute to a reduced dropout rate because participants might be less likely to drop out of treatment due to fewer logistical barriers to care and external stressors associated with being removed from their traditional environments. Although the inclusion of studies in which boarding was offered could have been a confounding variable, studies offering this option were included because staying overnight was not a requirement, and the ability to offer boarding/lodging can be viewed as a unique component of intensive treatments. Regular attendance at therapy appointments can be challenging, particularly for individuals who live in rural areas or cities in which access to providers with training in trauma-focused interventions and PTSD treatment is not readily available. Therefore, overnight lodging for treatment that is delivered over several days or weeks may be an advantage of intensive treatments (e.g., Bryan et al., 2018).

There is still much to learn regarding the impact of intensive ESTs for PTSD on symptom amelioration. The present findings highlight the potential benefits of increased session frequency on improvements in PTSD symptom severity and reduced treatment dropout. The variability of the included studies may complicate drawing definitive, generalizable conclusions; however, this variability also demonstrates ways in which ESTs for PTSD can be implemented flexibly among individuals who have experienced a range of trauma types.

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